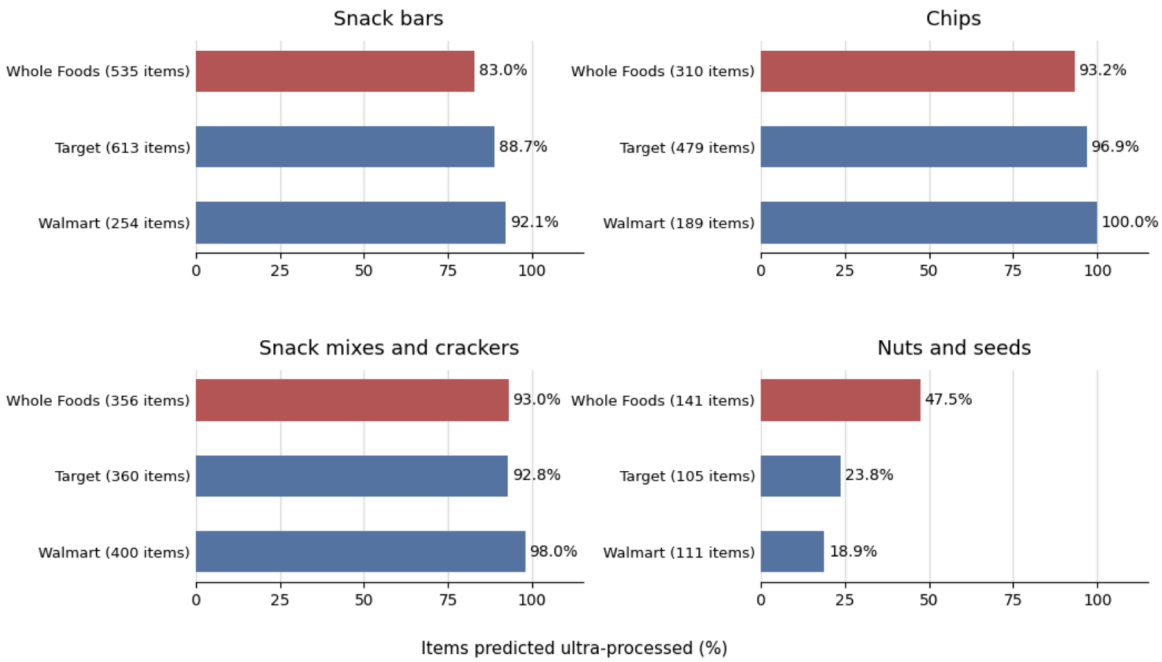
Does Whole Foods have less processed snacks than other grocery stores?



I wanted to see whether Whole Foods offers a lower ultra processed share than Target and Walmart as Whole Foods advertises that it avoids certain preservatives and artificial additives. The GroceryDB paper suggests that it often does, but a store-wide average could hide differences in particular foods. I chose snack categories because they are the food category with the most artificial ingredients and processed foods and choosing multiple snack categories would show whether the pattern was consistent, which it was not as Whole Foods had a lower predicted ultra processed share for bars and chips, 83% and 93.2% respectively, than either other store. However, Whole Foods had a higher share for nuts and seeds, 47.5% versus 23.8% at Target and 18.9% at Walmart. Whole Foods also had a higher share of ultra processed foods for snack mixes and crackers when compared to Target.

I kept the four snack categories with at least 100 listed products at each store, bars, chips, mixes and crackers, and nuts and seeds as they had a substantial sample size to test from. Target had only 12 dips and salsa products and 48 popcorn products, so I left those categories out rather than comparing small groups with hundreds of products at the other stores. For the remaining listings, I used the dataset's FPro class 3 label, which indicates that the item is ultra processed, and calculated its percentage separately for every store and category.

I used four small horizontal bar charts with a percentage scale as it makes store comparisons possible within a category while keeping the category names readable. I put stores in the same order in every panel, used red for Whole Foods and blue for the other two to show that Whole Foods was what is being investigated, and printed percentages at the bar ends so small differences near 100% are still visible. I also added the product counts for each store per category to show how many items exist for further context.

The charts do not display which specific products are being compared but are rather a reference for the snack options. More specifically, the nuts and seeds category also includes trail mixes which contain candy such as M&Ms. The chart also does not suggest that one store is healthier than the other.